

Statement of Purpose

I am a 3rd year B.Tech student in IT at VJTI (Veermata Jijabai Technological Institute). My interest in machine learning grew from watching what well built AI systems could do not just in research papers, but in real applications solving real problems. I wanted to understand how those systems were built from the inside, not just appreciate them from the outside. That curiosity is what pushed me to start learning about AI and ML.

My foundation became clear when I had a discrete mathematics course in which probability, statistics, linear algebra, and data structures were taught and that gave me the mathematical grounding to understand how machine learning works, not just how to run it. From there I moved to self-directed learning through coursera's courses like supervised and unsupervised learning and Deep learning specialization course. I completed all the courses and assignments given in deep learning course. It was where I first encountered neural networks, hyperparameter tuning and other optimization techniques. What kept me going was not the certificates, it was realising that the gap between understanding a concept and actually implementing it is where the real learning happens.

The project that defined my practical ML journey is RAGNexus, a full-stack Retrieval-Augmented Generation system I built from scratch. I designed a pipeline where documents are chunked, embedded using sentence-transformers, and stored in a Qdrant vector database. When a user asks a question, the system runs HyDE query expansion, followed by hybrid search combining dense vector retrieval and BM25 keyword matching, then cross-encoder reranking before passing top contexts to a Groq LLM for grounded answer generation with source citations. Building this taught me things no course covered why naive RAG fails, how retrieval quality directly caps generation quality, and how to think about evaluation when standard accuracy metrics don't apply. It also revealed exactly where my knowledge ends: I used pre-trained embedding models without fully understanding how they were trained, and I evaluated informally rather than rigorously. Those are the gaps I want to close.

What draws me to Amazon MLSS is the specific technical depth it offers in areas I am actively hitting walls in. I want to understand large-scale ML systems, how embeddings are trained, how retrieval systems are optimised at scale, and how generative AI is deployed responsibly in production. These are not things I can learn from documentation alone. Learning from Amazon engineers who have solved these problems at scale would give me something no courses can offer the reasoning behind architectural decisions. That kind of mentorship is exactly what my learning needs right now.

My long-term goal is to apply machine learning to problems where the stakes are real like healthcare, education, or climate where a better model is not just a benchmark improvement but something that changes an outcome for someone. MLSS would give me the depth to pursue that kind of work seriously.